

The trained circling rat: a model for inducing unilateral caudate dopamine metabolism

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Despite considerable knowledge about brain dopamine in drug-induced and pathological states, the dynamic relationship of dopamine to voluntary motor behaviour is not established. In an effort to produce selective, lateralized changes in brain dopamine metabolism associated with movement, we have now developed an animal model in which normal rats were trained to circle for a sucrose/water reward. Turning direction was randomly assigned. The effect of circling on dopamine metabolism was studied by killing animals during the reinforcement period and measuring caudate dopamine and dihydroxyphenylacetic acid (DOPAC) concentrations in both sides of brain. Prior to turning, caudate dopamine and DOPAC levels were the same bilaterally. By contrast, animals killed after 20 min of circling showed a 67% increase in dopamine and a 46% increase in DOPAC concentrations in the caudate contralateral to the turning direction while the ipsilateral caudate showed no concentration changes. Such a rapid, massive unilateral increase in caudate dopamine metabolism suggests that dopamine neurones in contralateral caudate are selectively activated during circling behaviour. This model should be useful for further study of the relationships between neurotransmitter activity, regional brain physiology and voluntary movement, without the need for drugs or brain lesions.

Circling behaviour is an extensively studied area of the neurosciences and has recently been reviewed¹. Studies have used lesions, stimulation of various brain areas and pharmacological manipulations in an effort to understand the neural basis of motor behaviour. Most workers have focused on the dopamine-containing nigrostriatal system. Ungerstedt showed that animals with unilateral 6-hydroxydopamine lesions of the substantia nigra would circle in the direction of the lesioned side after amphetamine administration, presumably because of enhanced dopamine release in the caudate on the intact side². In this lesion model, a receptor agonist such as apomorphine will produce turns in the direction opposite to the lesions, possibly because of the development of denervation supersensitivity on the lesioned side. Glick and co-workers have noted that some rats preferentially turn to one side spontaneously and that this turning behaviour can be increased by amphetamine^{3,4}. Following amphetamine, these animals show an increase in dopamine in the caudate contralateral to the turning direction. Results from these and other experiments have suggested that striatal dopamine has a role in circling behaviour. Because all of these paradigms have used drugs or lesions or animals with natural behavioural asymmetries, these previous studies do not directly address the relationship between dopamine metabolism and movement in the normal animal. We have therefore developed a behavioural model to study caudate dopamine metabolism in animals trained to turn in an arbitrary direction without the use of drugs or lesions.

Eighteen male Sprague-Dawley rats weighing 250–350 g were water deprived for 24 h before training. All rats were randomly assigned to be trained to turn 360° in the left-hand or right-hand direction. The training apparatus was a 24-cm diameter circular drum with 25-cm vertical walls. The drum was equipped with a liquid dispenser (Lafayette) calibrated to deliver 0.05 ml of a 10% w/w sucrose/water per reinforcement. Following water deprivation, rats were placed in the drum and sucrose solution was released from the dispenser. All rats quickly located the fluid. The training protocol consisted of rewarding each rat by successive approximations to a full turn in the prescribed direction. Initially, a quarter turn in the

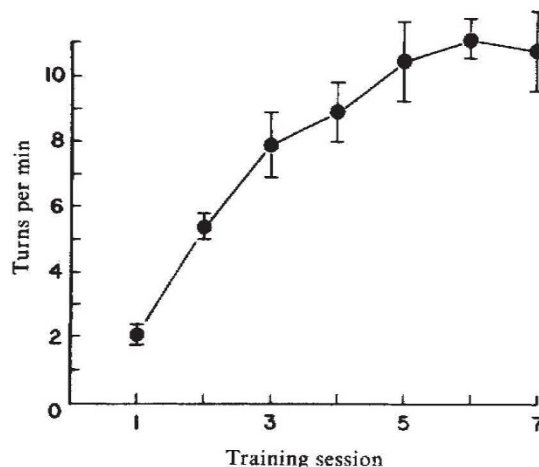


Fig. 1 Acquisition of trained turning behaviour in six animals. Water-deprived animals were trained daily using a sucrose/water reward. Animal performance plateaued after 7 days of reinforcement training.

appropriate direction was reinforced. When this response was acquired, a 180° turn was then necessary for reward. Subsequently, 270° and 360° turns were required. Rats learned this procedure within 20–30 min and the first training session continued until each animal had performed 100 turns. Training was conducted daily for 7 days according to a continuous

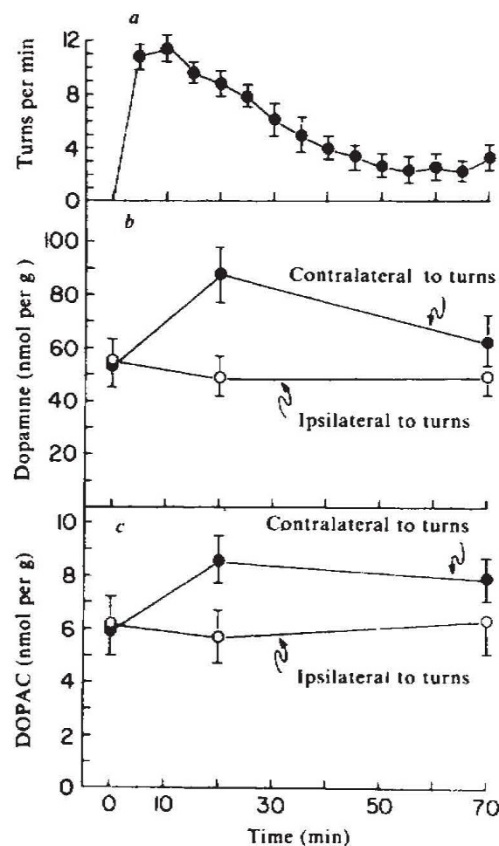


Fig. 2 Turning behaviour and caudate dopamine metabolism in trained rats. Trained animals were placed in the reinforcement apparatus at time 0. *a*, Turning began rapidly after water reinforcement was started. Data are mean $\pm$ s.e.m. for six animals. *b*, *c*, Caudate dopamine and DOPAC concentrations were measured by killing six animals at each time point. Values are mean $\pm$ s.e.m. Contralateral dopamine and DOPAC concentrations are significantly elevated compared with ipsilateral concentrations at 20 min ($P < 0.01$, both values) and 70 min ($P < 0.05$, both values) after starting circling. Concentrations are in nanomol per g wet weight caudate. Statistics were determined by Student's paired *t*-test.

reinforcement schedule. Figure 1 shows the acquisition curves for this behaviour. One hundred turns were required during the first 3 days of training while 150 complete turns were required for the subsequent 4 days of training. During this time, the only water provided was that received during the training session. *Ad libitum* access to Purina Laboratory Chow was available in the home cage. This deprivation schedule enabled all rats to maintain at least 85% of their original body weight.

We found that after turning behaviour was acquired, the animals retained the response for at least 1 month without regular training sessions. Minimal training was required to restore the previous maximum turning intensity. After animals had learned the turning paradigm, they were killed at different times during circling behaviour and caudate dopamine and DOPAC measured. Rats were killed by decapitation, and left and right caudate were dissected and assayed for dopamine and DOPAC by HPLC with electrochemical detection, using trichloroacetic acid as the ion-pairing agent according to the method of Freed and Asmus⁵.

The relationship between turning and dopamine metabolism is presented in Fig. 2. Figure 2a shows the time course of turning for animals permitted to follow the full turning protocol. No circling behaviour occurred during the baseline period before water reinforcement was made available. When the animals were transferred to the reinforcement apparatus, turning began immediately with high intensity. Turning rate declined after ~30 minutes and approached a low frequency plateau by 70 min. Caudate nuclei contralateral and ipsilateral to the trained circling direction were analysed for dopamine and its major metabolite, DOPAC (see Fig. 2b, c). Dopamine and DOPAC concentrations significantly increased in the caudate contralateral to the trained turning direction at the time of peak turning intensity. Contralateral concentrations were increased both in comparison with baseline values and with simultaneous values from the ipsilateral side. After 70 min of reinforcement, circling had slowed and contralateral dopamine and DOPAC concentrations declined but were still significantly greater than ipsilateral concentrations.

Our findings of unilateral increases in both dopamine and DOPAC concentrations imply that both synthesis and turnover of caudate dopamine are selectively enhanced during voluntary turning. Previous studies of dopamine metabolism in brain of resting animals have suggested that there is a small, readily releasable high turnover pool of dopamine in the caudate and a large slow turnover storage pool⁶. Our results indicate that the trained circling rat is capable of unilaterally increasing total caudate dopamine concentration by 67% in 20 min. Although we have not localized this dopamine, such a massive increase suggests that the entire dopamine pool is involved. Because contralateral caudate DOPAC concentration is also elevated over the time course of circling, increased dopamine synthesis is accompanied by increased transmitter turnover. Dopamine concentration rose more rapidly than DOPAC concentration (67% increase in dopamine versus 46% for DOPAC at 20 min), but at 70 min, as turning subsided, dopamine concentration had fallen to only 20% greater than baseline while DOPAC remained 35% increased. As DOPAC is a metabolite of dopamine, this delayed fall-off of DOPAC concentration is expected.

We have obtained further support for asymmetric regulation of caudate dopamine metabolism in independent experiments. Using the trained rat model and the technique of *in vivo* electrochemistry, we have found that contralateral release of dopamine is relatively increased during circling⁷. Furthermore, we have assayed tyrosine hydroxylase activity at the same time points reported here using the method of Nagatsu⁸ at a sub-saturating concentration of pterin cofactor (0.4 mM 2-amino-4-hydroxy-6-methyltetrahydropterin). Baseline activity was not significantly different in the two sides of brain (ratio of activity contralateral/ipsilateral = 0.84 ± 0.08 , $n = 5$). After 20 min circling, contralateral caudate tyrosine hydroxylase activity increased 106% relative to the ipsilateral side (activity ratio

1.73 ± 0.21 , $n = 5$, $P < 0.01$ compared with baseline ratio). These data are part of a complete study of the kinetics of tyrosine hydroxylase activation in our circling rat model (M.E. Morgan, B.K.Y. and C.R.F., in preparation).

Ours is the first observation of lateralization of brain neurotransmitter metabolism during voluntary motor behaviour. Nieoullon *et al.* demonstrated enhanced dopamine release in caudate ipsilateral to unilateral sensory stimulation in anaesthetized cats⁹. In addition, their experiments showed a reduction in dopamine release from the contralateral caudate. Glick and co-workers showed relative increases in contralateral caudate dopamine concentration after amphetamine administration to unlesioned animals which showed side preferences³, but our experiments are the first evidence that such changes occur in normal animals executing an arbitrarily assigned behaviour. Furthermore, the time course of the changes in dopamine concentration parallel the motor behaviour. Our experiments do not address the question of the causal role of dopamine in movement. Although the maximum increase in contralateral dopamine concentration is observed at the time of maximum circling intensity and is reduced as turning subsequently slows, we cannot say whether dopamine has caused or is simply the result of the turning behaviour.

The traditional substantia nigra-lesioned rotating rat model has been useful for investigating nigrostriatal function and as a screening device for potential anti-parkinsonian drugs. However, the trained rat model has an important advantage for studying neurotransmitters in the central nervous system. It provides a method for exploring the relationships between behaviour and dopamine metabolism in the normal, undrugged and unlesioned brain. In addition, the trained turning rat may be useful for studying laterality in other brain areas linked to motor function and to other neurotransmitters such as γ -aminobutyric acid and acetylcholine.

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Brain grafts reverse hypogonadism of gonadotropin releasing hormone deficiency

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Hypogonadism in the mutant *hpg* mouse is characterized by a deficiency of hypothalamic gonadotropin releasing hormone (GnRH)¹. Affected male mice exhibit immature reproductive organs, small abdominal testes and low pituitary and plasma gonadotropin concentrations. Recent studies have demonstrated the potential of fetal brain transplants to establish functional connections with host tissues². We therefore sought to use this approach to correct the *hpg* deficit. Fetal preoptic area (POA) (a site of GnRH production) from unaffected animals of the *hpg* strain was transplanted into the anterior third ventricle of adult *hpg* mice. We report that in such implanted

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